

Supplementary Online Content

Ambalavanan N, Carlo WA, Nowak KJ, et al; *Eunice Kennedy Shriver* National Institute of Child Health and Human Development Neonatal Research Network. Early intratracheal budesonide to reduce bronchopulmonary dysplasia in extremely preterm infants: the Budesonide in Babies (BiB) randomized clinical trial. *JAMA*. Published online September 30, 2025. doi:10.1001/jama.2025.16450

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This supplementary material has been provided by the authors to give readers additional information about their work.

eTable 1. Analysis Populations and Participant Disposition

	ITT^a (n=641)	ITT excluding untreated participants^b (n=635)	SAF^c (n=635)	PP^d (n=617)
Treated, n (%)	635 (99.1)	635 (100.0)	635 (100.0)	617 (100.0)
Treatment Group, n (%)	As randomized	As randomized	As treated	As treated
Budesonide + Surfactant	323 (50.4)	320 (50.4)	322 (50.7)	309 (50.1)
Surfactant Alone	318 (49.6)	315 (49.6)	313 (49.3)	308 (49.9)
Primary Endpoint Status (36 weeks PMA), n (%)				
Completed	639 (99.7)	634 (99.8)	634 (99.8)	617 (100.0)
Ended Early	2 (0.3)	1 (0.2)	1 (0.2)	0 (0.0)
GDB status (up to 120 days postnatal age), n (%)				
Discharged home	340 (53.9)	337 (53.7)	337 (53.7)	329 (53.8)
Still in hospital at 120 days	189 (30.0)	189 (30.1)	189 (30.1)	184 (30.1)
Transferred to another facility	16 (2.5)	16 (2.5)	16 (2.5)	15 (2.5)
Death	86 (13.6)	86 (13.7)	86 (13.7)	84 (13.7)

CA = corrected age; GDB = Generic Database; ITT = intention-to-treat; PMA = postmenstrual age; PP = per-protocol; SAF = safety.

^a The ITT population includes all randomized infants, grouped by randomized treatment assignment.

^b The ITT excluding untreated participants population is comprised of all randomized infants who received at least one dose of study drug.

^c The SAF population includes all infants who received at least one dose of study drug, grouped by actual treatment received; infants who received budesonide for either dose 1 or dose 2 are counted as Budesonide + Surfactant arm.

^d The PP population is comprised of all randomized patients who received at least one dose of study drug and did not experience any major protocol violations. See Figure 1 for additional information.

eTable 2. Extent of Exposure, Safety Population

	Budesonide + Surfactant (n=322)	Surfactant Alone (n=313)
<i>Study Drug Exposures</i>		
Number of Doses, n (%)		
One Dose	220 (68.3)	195 (62.3)
Two Doses	102 (31.7)	118 (37.7)
Postnatal age (h) of Dose 1	(n=322)	(n=313)
Mean (SD)	4.05 (7.47)	3.70 (6.80)
Median (Q1-Q3)	1.7 (0.5 - 3.6)	1.6 (0.6 - 3.3)
Min-Max	0.0 - 48.4	0.0 - 49.2
Postnatal age (h) of Dose 2	(n=102)	(n=118)
Mean (SD)	23.09 (11.81)	21.98 (10.12)
Median (Q1-Q3)	17.8 (13.7 - 29.7)	18.6 (14.3 - 25.9)
Min-Max	7.6 - 59.9	9.5 - 49.4
<i>Study Drug Compliance</i>		
Not treated per-protocol ^a , n (%)	36 (11.2)	29 (9.3)
Group assignment violation, n (%)	3 (0.9)	1 (0.3)
Dose volume violation ^b , n (%)	31 (9.6)	27 (8.6)
Administration violation, n (%)	2 (0.6)	1 (0.3)
Surfactant volume compliance (%) for Dose 1 ^{b, c}	(n=322)	(n=313)
Mean (SD)	99.35 (5.22)	100.03 (4.73)
Median (Q1-Q3)	100.0 (99.6 - 100.3)	100.0 (99.8 - 100.5)
Min-Max	49.9 - 120.0	51.2 - 120.0
Surfactant volume compliance (%) for Dose 2 ^{b, c}	(n=102)	(n=118)
Mean (SD)	102.12 (14.71)	101.70 (10.67)
Median (Q1-Q3)	100.0 (99.1 - 101.5)	100.1 (99.6 - 101.6)
Min-Max	80.0 - 201.0	74.1 - 191.3
Budesonide volume compliance (%) for Dose 1 ^{b, c}	(n=319)	NA
Mean (SD)	102.29 (24.06)	NA
Median (Q1-Q3)	100.0 (100.0 - 100.0)	NA
Min-Max	25.0 - 400.0	NA
Budesonide volume compliance (%) for Dose 2 ^{b, c}	(n=102)	NA

	Budesonide + Surfactant (n=322)	Surfactant Alone (n=313)
Mean (SD)	99.76 (7.83)	NA
Median (Q1-Q3)	100.0 (100.0 - 100.0)	NA
Min-Max	25.0 - 109.9	NA

h = hour; NA = not applicable; Q1 = first quartile (25th percentile); Q3 = third quartile (75th percentile); SD = standard deviation.

The safety population includes infants who received at least one dose of study intervention, and treatment group is reported as treatment received. Infants who were randomized but not treated are excluded from this analysis population.

^a Infants not treated per protocol received at least one dose of study drug with a dosing violation. Dosing violations included treatment under a different group than randomized, an incorrect dose volume by weight, and/or non-compliant physical preparation or administration of study drug. Infants could have multiple reasons for one or both doses being non-compliant.

^b Dosing volume compliance is reported as the percent of drug volume administered, relative to the expected drug volume calculated by body weight (i.e., 100% indicates a dose volume at the exact, per-protocol volume). Volume compliance is assessed separately for the surfactant and budesonide components of each dose, and volume compliance is independent of group assignment (e.g., an infant randomized to surfactant alone but received budesonide can have a compliant budesonide volume by weight). Dosing volumes within +/- 10% of the per-protocol dose (i.e., 90% - 110%) were considered protocol compliant.

^c The surfactant per protocol dose was 2.5 mL/kg by weight for Dose 1 and 1.25 mL/kg for Dose 2. The budesonide per protocol dose was 1 mL/kg by weight for Dose 1 and Dose 2, then mixed with the dose-specific surfactant volume.

eTable 3. Adverse Event Experience, Safety Population

Safety Outcome	Budesonide + Surfactant (n=322)	Surfactant Alone (n=313)	Relative Risk (95% CI) or P-value
<i>Adverse Events</i>			
Experienced Any AEs, n (%)	242 (75.4)	202 (64.5)	1.16 (1.05, 1.28)
Any of interest ^a , n (%)	240 (74.8)	193 (61.7)	1.20 (1.09, 1.33)
Any AEs by Category, n (%)			
Endotracheal Tube Blockage ^b	3 (0.9)	4 (1.3)	0.73 (0.16, 3.28)
Prolonged Hypoxemia + Bradycardia ^b	2 (0.6)	3 (1.0)	0.65 (0.11, 3.90)
Pulmonary Air Leak	16 (5.0)	18 (5.8)	0.86 (0.45, 1.66)
Hypotension	56 (17.4)	59 (18.8)	0.92 (0.67, 1.27)
Hypertension	28 (8.7)	21 (6.7)	1.28 (0.75, 2.10)
Hyperglycemia	214 (66.7)	156 (49.8)	1.33 (1.17, 1.51)
Intracranial Hemorrhage	40 (12.5)	37 (11.8)	1.03 (0.69, 1.54)
Early-Onset Sepsis	9 (2.8)	8 (2.6)	1.06 (0.41, 2.75)
Late-Onset Sepsis	70 (22.4)	70 (23.7)	0.93 (0.70, 1.24)
Periventricular Leukomalacia	15 (4.8)	13 (4.3)	1.09 (0.52, 2.26)
Spontaneous Intestinal Perforation	17 (5.3)	9 (2.9)	1.82 (0.83, 4.02)
Other (specify) ^a	27 (8.4)	32 (10.2)	0.80 (0.50, 1.30)
Pulmonary Hemorrhage	11 (3.4)	14 (4.5)	0.75 (0.35, 1.62)
AE Burden (#AE/participant) ^c			
Mean (SD)	1.6 (1.5)	1.3 (1.4)	P = 0.004
Median (IQR)	1 (1 - 2)	1 (0 - 2)	NA
<i>Serious Adverse Events</i>			
Experienced Any SAEs, n (%)	64 (19.9)	54 (17.3)	1.13 (0.82, 1.55)
Any of interest ^a , n (%)	53 (16.5)	42 (13.4)	1.20 (0.83, 1.73)
Any fatal ^d , n (%)	22 (6.9)	25 (8.0)	0.84 (0.49, 1.44)
Any SAEs by Category, n (%)			
Endotracheal Tube Blockage ^b	1 (0.3)	1 (0.3)	0.98 (0.06, 15.66)
Prolonged Hypoxemia + Bradycardia ^b	1 (0.3)	2 (0.6)	0.49 (0.04, 5.39)
Pulmonary Air Leak ^e	5 (1.6)	12 (3.8)	0.37 (0.13, 1.06)
Hypotension	13 (4.0)	8 (2.6)	1.53 (0.64, 3.63)
Hypertension ^b	0 (0.0)	1 (0.3)	NA

Safety Outcome	Budesonide + Surfactant (n=322)	Surfactant Alone (n=313)	Relative Risk (95% CI) or P-value
Hyperglycemia ^b	8 (2.5)	3 (1.0)	2.64 (0.69, 10.05)
Intracranial Hemorrhage	17 (5.3)	19 (6.1)	0.85 (0.45, 1.58)
Early-Onset Sepsis ^b	4 (1.2)	2 (0.6)	1.96 (0.36, 10.79)
Late-Onset Sepsis ^b	5 (1.6)	2 (0.6)	2.46 (0.47, 12.78)
Periventricular Leukomalacia	0 (0.0)	0 (0.0)	NA
Spontaneous Intestinal Perforation	15 (4.7)	9 (2.9)	1.61 (0.72, 3.61)
Other (specify) ^a	18 (5.6)	20 (6.4)	0.84 (0.46, 1.55)
Pulmonary Hemorrhage	8 (2.5)	8 (2.6)	0.94 (0.35, 2.48)
SAE Burden (#SAE/participant) ^c			
Mean (SD)	0.3 (0.6)	0.3 (0.6)	P = 0.42
Median (Q1-Q3)	0 (0 - 0)	0 (0 - 0)	NA

AE = adverse event; CI = confidence interval; NA = not applicable; P = p-value; Q1 = first quartile (25th percentile); Q3 = third quartile (75th percentile); SAE = serious adverse event; SD = standard deviation.

Adverse events were monitored through 7 days/168 hours after last dose of study drug, with the exception of Spontaneous Intestinal Perforation (SIP) and Periventricular Leukomalacia (PVL) which were monitored through 30 days/720 hours after last dose of study drug. Non-serious adverse events were followed until the end of the respective monitoring window; serious adverse events were followed until resolution or study status. Serious adverse events were adverse events deemed fatal, life threatening, or to require medical or surgical intervention to prevent death, a life threatening event, prolonged hospitalization, and/or persistent or significant disability or incapacity.

The safety population includes infants who received at least one dose of study intervention, and treatment group is reported as treatment received. Infants who were randomized but not treated are excluded from this analysis population.

Relative risk and 95% confidence intervals were estimated for the Budesonide + Surfactant group, relative to the Surfactant Alone group. Treatment groups were compared using robust Poisson regression, adjusting for gestational age strata and pooled center, unless otherwise noted. All analyses were performed at the alpha = 0.05 significance level for descriptive purposes only.

^a Adverse events of interest were pre-specified in the protocol and are listed as event categories in this table. All other adverse events occurring within 7 days/168 hours after last dose of study drug are reported as the Other (Specify) category. Pulmonary hemorrhage was the most prevalent event within the Other (Specify) category.

^b Relative risk was approximated by the crude (unadjusted) odds ratio with exact 95% confidence intervals due to low event prevalence.

^c Wilcoxon rank sum tests were used to compare event burden outcomes across arms. P-values are reported; relative risks are not applicable.

^d Fatal adverse events include all deaths within 7 days/168 hours after last dose of study drug, regardless of event category. Adverse events that became serious within the monitoring window (30 days for SIP and PVL; 7 days otherwise) and resulted in death by end of study (36 weeks postmenstrual age) are also considered fatal adverse events.

^e Relative risk was estimated by Mantel-Haenszel methods, approximated by the common odds ratio, due to model convergence issues. The analysis was stratified by the cross-classifications of gestational age strata and pooled center.

eTable 4. Clinical and Growth Outcomes, Safety Population

Safety Outcome	Budesonide + Surfactant (n=322)	Surfactant Alone (n=313)	Relative Risk (RR) or Mean Difference (MD) (95% CI)
<i>Clinical Outcomes by 120d PNA ^a</i>			
Death before 120 days PNA, n/N (%)	50/321 (15.6)	44/313 (14.1)	RR: 1.09 (0.76, 1.57)
Median (Q1-Q3)	12 (5 - 33)	10 (2 - 18)	
Min-Max	1 - 89	1 - 115	
NEC or Death, n/N (%)	77/320 (24.1)	72/313 (23.0)	RR: 1.04 (0.79, 1.37)
NEC, n/N (%)	35/319 (11.0)	40/308 (13.0)	RR: 0.85 (0.56, 1.29)
PDA or Death, n/N (%)	182/320 (56.9)	197/313 (62.9)	RR: 0.89 (0.79, 1.00)
PDA, n/N (%)	159/319 (49.8)	175/308 (56.8)	RR: 0.86 (0.75, 0.99)
PDA managed with medical therapy or Death, n/N (%)	121/316 (38.3)	128/311 (41.2)	RR: 0.91 (0.76, 1.09)
PDA managed with medical therapy, n/N (%)	84/315 (26.7)	98/306 (32.0)	RR: 0.80 (0.64, 1.01)
PDA managed with surgery or cardiac catheterization or Death, n/N (%)	54/320 (16.9)	47/313 (15.0)	RR: 1.11 (0.78, 1.57)
PDA managed with surgery or cardiac catheterization ^b , n/N (%)	5/319 (1.6)	3/308 (1.0)	RR: 1.62 (0.38, 6.83)
Stage 3 ROP or Death, n/N (%)	87/318 (27.4)	83/310 (26.8)	RR: 1.02 (0.80, 1.29)
Stage 3 ROP, n/N (%)	37/277 (13.4)	40/270 (14.8)	RR: 0.87 (0.60, 1.28)
<i>Growth Outcomes at 36w PMA ^e</i>			
Weight Z-score, Mean (SD)	-1.1 (0.9)	-1.2 (0.8)	MD: 0.07 (-0.08, 0.22)
<10th percentile Weight, n/N (%)	107/261 (41.0)	113/259 (43.6)	RR: 0.95 (0.78, 1.15)
<10th percentile Weight or Death, n/N (%)	155/309 (50.2)	154/300 (51.3)	RR: 0.98 (0.84, 1.14)
Length Z-score, Mean (SD)	-2.0 (1.0)	-2.0 (1.0)	MD: 0.04 (-0.12, 0.21)
<10th percentile Length, n/N (%)	201/261 (77.0)	201/259 (77.6)	RR: 0.99 (0.90, 1.08)
<10th percentile Length or Death, n/N (%)	249/309 (80.6)	242/300 (80.7)	RR: 1.00 (0.93, 1.08)
Head circumference Z-score, Mean (SD)	-1.5 (1.0)	-1.6 (1.0)	MD: 0.04 (-0.132, 0.203)
<10th percentile Head circumference, n/N (%)	141/261 (54.0)	140/258 (54.3)	RR: 1.00 (0.85, 1.16)
<10th percentile Head circumference or Death, n/N (%)	189/309 (61.2)	181/299 (60.7)	RR: 1.01 (0.89, 1.14)

CI = confidence interval; d = day; MD = mean difference; NEC = necrotizing enterocolitis; PDA = patent ductus arteriosus; PMA = postmenstrual age; PNA = postnatal age; Q1 = first quartile (25th percentile); Q3 = third quartile (75th percentile); ROP = retinopathy of prematurity; RR = risk difference; w = week.

The safety population includes infants who received at least one dose of study intervention, and treatment group is reported as treatment received. Infants who were randomized but not treated are excluded from this analysis population. n/N (%) reported for each outcome is the number of participants experiencing the outcome over the number of participants with a nonmissing value for the outcome.

Binary outcomes report relative risks, estimated by robust Poisson regression, unless otherwise noted. Continuous outcomes report least squares mean difference estimated by robust generalized linear models (normal distribution with identity link). Both model types compared the Budesonide + Surfactant group to the Surfactant Alone group, adjusting for gestational age strata and pooled center. All analyses were performed at the alpha = 0.05 significance level for descriptive purposes only.

^a Clinical outcomes are assessed through Generic Database status (120 days postnatal age and in hospital, or prior death, transfer, or discharge to home). NEC, PDA, and ROP outcomes are not available for infants who died within 12 hours postnatal age. Death composites for clinical outcomes consider in-hospital deaths by 120d PNA.

^b Relative risk was approximated by the crude (unadjusted) odds ratio with exact 95% confidence intervals due to low event prevalence.

^c Growth outcomes are collected at 36 weeks PMA and are not available for infants who reach Generic Database status prior to 35 weeks PMA. Growth measurements were standardized to the Olsen et al. (2010) growth curves for United States data. Death composites for growth outcomes consider in-hospital deaths by 36w PMA.

eTable 5. Protocol Deviation and Violation Experience, Intention-to-Treat Population

Participants with Non-compliances	Budesonide + Surfactant (n = 323)	Surfactant Alone (n = 318)
<i>Any protocol deviation/violation, n (%)</i>	69 (21.4)	51 (16.0)
Any major deviation/violation ^a	11 (3.4)	7 (2.2)
<i>Protocol violation by type, n (%)</i>		
Lack of consent ^b	2 (0.6)	0 (0.0)
Ineligible enrollment ^c	3 (0.9)	4 (1.3)
Randomized but never initiated study drug ^d	3 (0.9)	3 (0.9)
Randomized to wrong GA strata	8 (2.5)	8 (2.5)
Randomized > 48 hours PNA	0 (0.0)	1 (0.3)
Received wrong study drug (treatment group) - Dose 1	1 (0.3)	0 (0.0)
Received wrong study drug (treatment group) - Dose 2	0 (0.0)	3 (0.9)
Received incorrect dose volume/concentration - Dose 1	24 (7.4)	17 (5.3)
Received incorrect dose volume/concentration - Dose 2	9 (2.8)	13 (4.1)
Infant received study drug > 50 hours PNA - Dose 2	1 (0.3)	0 (0.0)
Preparation/administration of study drug NPP - Dose 1	2 (0.6)	1 (0.3)
Preparation/administration of study drug NPP - Dose 2	0 (0.0)	0 (0.0)
Unintentional unmasking	6 (1.9)	1 (0.3)
Indomethacin use during 7 days after last study drug dose	1 (0.3)	3 (0.9)
Other violations - Dose 1	1 (0.3)	0 (0.0)
Other violations - Dose 2	0 (0.0)	0 (0.0)
Other violations - Non-Dose	5 (1.5)	1 (0.3)
<i>Protocol deviation by type, n (%)</i>		
Infant aged-out of randomized GA strata	0 (0.0)	0 (0.0)
Open label surfactant administered instead of intended second dose of study drug	7 (2.2)	4 (1.3)
BPD assessment not administered per protocol	8 (2.5)	3 (0.9)
Other deviation - Dose 1	0 (0.0)	0 (0.0)
Other deviation - Dose 2	1 (0.3)	1 (0.3)
Other deviation - Non-Dose	4 (1.2)	2 (0.6)

BPD = bronchopulmonary dysplasia; GA = gestational age; NPP = not per-protocol; PNA = postnatal age.

Most deviation/violation types occurred once per participant. As such, the participant experience roughly corresponds to total event frequency.

^a Participants with major protocol deviations/violations are excluded from the per-protocol analysis population. Major violations include treatment without consent, treatment while ineligible, treatment under the wrong arm or both arms, treatment after the dosing window, infants receiving less than half or more than 2.4 times the intended dose, and dosing administration violations. All participants with major deviations/violations have corresponding protocol deviations; however, the major event may be captured under multiple categories and/or the 'Other' category.

^b The two consent violations reported in this table were randomized and treated without affirmed consent. One declined consent post-treatment and was subsequently withdrawn from the study; the other consented post-treatment and proceeded with the study. A third infant was randomized and treated without consent (not shown). The parent(s)/guardian(s) declined consent postnatally and requested their data be excluded from the study. This infant is captured in the CONSORT diagram but excluded from all other analyses and displays.

^c This study defines enrollment at time of randomization. Four infants were assessed to be eligible at the time of randomization but were later deemed ineligible; two ineligible infants were randomized in error but were not treated since they were known to be ineligible; one infant was randomized and treated in error.

^d Four infants became ineligible for study participation after randomization and were not treated (two infants' health improved and did not need surfactant, two infants' health declined with one infant dying and another receiving comfort care); two ineligible infants were randomized in error but were not treated since they were known to be ineligible (see above footnote).

eTable 6. Summary of Adverse Events, Safety Population

Adverse Events	Budesonide + Surfactant (n=322)	Surfactant Alone (n=313)
<i>Number of events, n</i>	500	403
Events of Interest ^a	468	366
Unexpected Events ^b	83	65
<i>By Category ^a, n</i>		
Endotracheal Tube Blockage	3	4
Prolonged Hypoxemia And Bradycardia	2	3
Pulmonary Air Leak	17	20
Hypotension	61	63
Hypertension	31	23
Hyperglycemia	272	193
Intracranial Hemorrhage	40	37
Early-Onset Sepsis	7	5
Late-Onset Sepsis	13	7
Periventricular Leukomalacia	5	2
Spontaneous Intestinal Perforation	17	9
Other (specify) ^a	32	37
Pulmonary Hemorrhage	11	14
Pulmonary Hypertension	2	3
Persistent Pulmonary Hypertension	1	2
Respiratory Failure	3	2
Respiratory Distress Syndrome	3	3
<i>By Severity, n</i>		
Mild	7	6
Moderate	234	199
Severe	188	128
Life Threatening	43	44
Fatal	28	26
<i>By Relationship to Treatment, n</i>		
Not Related	124	105
Unlikely Related	236	209

Adverse Events	Budesonide + Surfactant (n=322)	Surfactant Alone (n=313)
Possibly Related	135	82
Probably Related	5	7
<i>By Action Taken for Treatment, n</i>		
No change	499	402
Dose interrupted	1	1
<i>By Outcome^c, n</i>		
Resolved	273	209
Resolved w/ sequelae	4	5
Resolving	71	60
Not resolved	124	103
Fatal	28	26

Adverse events were monitored through 7 days/168 hours after last dose of study drug, with the exception of spontaneous intestinal perforation (SIP) and periventricular leukomalacia (PVL) which were monitored through 30 days/720 hours after last dose of study drug. Non-serious adverse events were followed until the end of the respective monitoring window; serious adverse events were followed until resolution or study status. Serious adverse events were adverse events deemed fatal, life threatening, or to require medical or surgical intervention to prevent death, a life threatening event, prolonged hospitalization, and/or persistent or significant disability or incapacity.

The safety population includes infants who received at least one dose of study intervention, and treatment group is reported as treatment received. Infants who were randomized but not treated are excluded from this analysis population.

This table summarizes the number of events observed in each treatment group; column headers provide the total number of participants in the treatment group for reference.

^a Adverse events of interest were pre-specified in the protocol and are listed as event categories in this table. All other adverse events occurring within 7 days/168 hours after last dose of study drug are reported in the Other (specify) category. Other event types that were observed at least 3 times are summarized as subcategories of the Other (specify) adverse event category.

^b Unexpected adverse events were of a nature, severity, or frequency that was not consistent with the known or foreseeable risks associated with the treatment agent(s) or the expected natural progression of any pre-existing disease, disorder, or condition.

^c Event outcome reflects resolution status at the end of adverse event monitoring. Non-serious adverse events were monitored for onset and resolution through 7 days/168 hours after last dose of study drug (30 days/720 hours for SIP and PVL). Events that became serious during the safety monitoring window were followed through resolution or primary study status (36 weeks' post-menstrual age), whichever came first.

eTable 7. Summary of Serious Adverse Events, Safety Population

Serious Adverse Events	Budesonide + Surfactant (n=322)	Surfactant Alone (n=313)
<i>Number of events, n</i>	90	80
Events of Interest ^a	70	59
Unexpected Events ^b	27	24
<i>By Category ^a, n</i>		
Endotracheal Tube Blockage	1	1
Prolonged Hypoxemia And Bradycardia	1	2
Pulmonary Air Leak	5	12
Hypotension	14	8
Hypertension	0	1
Hyperglycemia	8	3
Intracranial Hemorrhage	17	19
Early-Onset Sepsis	4	2
Late-Onset Sepsis	5	2
Spontaneous Intestinal Perforation	15	9
Other (specify) ^a	20	21
Pulmonary Hemorrhage	8	8
Pulmonary Hypertension	2	1
Persistent Pulmonary Hypertension	1	1
Respiratory Failure	2	2
Respiratory Distress Syndrome	3	3
<i>By Severity, n</i>		
Severe	19	10
Life Threatening	43	44
Fatal	28	26
<i>By Relationship to Treatment, n</i>		
Not Related	28	23
Unlikely Related	45	44
Possibly Related	17	12
Probably Related	0	1
<i>By Action Taken for Treatment, n</i>		

Serious Adverse Events	Budesonide + Surfactant (n=322)	Surfactant Alone (n=313)
No change	90	79
Dose interrupted	0	1
By Outcome^c, n		
Resolved	19	14
Resolved w/ sequelae	3	1
Resolving	10	11
Not resolved	30	28
Fatal	28	26

Adverse events were monitored through 7 days/168 hours after last dose of study drug, with the exception of Spontaneous Intestinal Perforation (SIP) and Periventricular Leukomalacia (PVL) which were monitored through 30 days/720 hours after last dose of study drug. Non-serious adverse events were followed until the end of the respective monitoring window; serious adverse events were followed until resolution or study status. Serious adverse events were life threatening or fatal or required medical or surgical intervention to prevent death, a life threatening event, prolonged hospitalization, and/or persistent or significant disability or incapacity.

The safety population includes infants who received at least one dose of study intervention, and treatment group is reported as treatment received. Infants who were randomized but not treated are excluded from this analysis population.

This table summarizes the number of events observed in each treatment group; column headers provide the total number of participants in the treatment group for reference.

^a Adverse events of interest were pre-specified in the protocol and are listed as event categories in this table. All other adverse events occurring within 7 days/168 hours after last dose of study drug are reported as the Other (Specify) category. Other event types that were observed at least 3 times are summarized as subcategories of the Other (specify) adverse event category.

^b Unexpected adverse events were of a nature, severity, or frequency that was not consistent with the known or foreseeable risks associated with the treatment agent(s) or the expected natural progression of any pre-existing disease, disorder, or condition.

^c Event outcome reflects resolution status at the end of adverse event monitoring. Non-serious adverse events were monitored for onset and resolution through 7 days/168 hours after last dose of study drug (30 days/720 hours for SIP and PVL). Events that became serious during the safety monitoring window were followed through resolution or primary study status (36 weeks post-menstrual age), whichever came first.

eTable 8. In-hospital Deaths by 120 Days' Postnatal Age, Safety Population

Death Details	Budesonide + Surfactant (n=322)	Surfactant Alone (n=313)
Died by 120 days PNA, n/N (%)	50/321 (15.6)	44/313 (14.1)
Died by 36 weeks PMA, n/N (%)	48/321 (15.0)	41/313 (13.1)
Died by 7 days PLD ^a , n/N (%)	18/321 (5.6)	21/313 (6.7)
Postnatal Age at Death (days)	(n=50)	(n=44)
Mean (SD)	21.9 (22.8)	18.0 (26.3)
Median (Q1-Q3)	12 (5 - 33)	10 (2 - 18)
Min-Max	1 - 89	1 - 115
Any Fatal SAEs ^a , n/N (%)	22/321 (6.9)	25/313 (8.0)
Primary Cause of Death ^b , n		
RDS	5	4
RDS with severe intracranial hemorrhage	3	2
RDS with infection	1	3
RDS with massive pulmonary hemorrhage	6	4
BPD	0	2
PPHN	1	1
Suspect sepsis/infection	5	1
Proven sepsis/infection	12	2
NEC	5	4
NEC with sepsis	1	4
Spontaneous perforation	0	2
Severe intracranial hemorrhage	0	2
Severe IVH with infection	2	0
Renal failure	1	1
Other	8	12

BPD = bronchopulmonary dysplasia; IVH = intraventricular hemorrhage; NEC = necrotizing enterocolitis; PPHN = persistent pulmonary hypertension; PLD = post last dose of study drug; PMA = postmenstrual age; PNA = postnatal age; Q1 = first quartile (25th percentile); Q3 = third quartile (75th percentile); RDS = respiratory distress syndrome; SAE = serious adverse event; SD = standard deviation; w = weeks'.

This table reports all-cause, in-hospital deaths through 120 days postnatal age. The censoring timepoint for the Neonatal Research Network Generic Data Base is 120 days PNA, the primary study endpoint for the BiB protocol is assessed at 36w PMA, and the main safety monitoring window for the BiB protocol is 7 days/168 hours after last dose of study drug.

The safety population includes infants who received at least one dose of study intervention, and treatment group is reported as treatment received. Infants who were randomized but not treated are excluded from this analysis population.

^a All fatalities within 7 days/168 hours after last dose of study drug are reported as fatal SAEs. All fatal spontaneous intestinal perforation or periventricular leukomalacia adverse events within 30 days/720 hours after last dose of study drug are reported as fatal SAEs. Any fatal SAEs outside of these windows resulted from events that became serious within the respective monitoring window.

^b Primary cause of death was sourced from the Neonatal Research Network Generic Data Base. These values may include autopsy results and may differ from the fatal adverse event term(s).